Meghna Banerjee

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Experience

- **Margadeshaka** — Chandigarh, India
  AI Intern — *Oct 2026 – Present*
  - **Working on an AI-driven healthcare assistant**: contributing to the design of an agent that organizes medical records and helps users with next steps.
  - **Defining the problem and system approach**: documenting the idea as a hypothesis, outlining assumptions, success criteria, and the planned workflow before development.
  - **Following a structured development process**: preparing behavior-driven test cases, architectural decision records, and proof-of-concept plans.

Projects

- **Airnexa** | Link — *Java, HTML/CSS, MySQL*
  - Built a flight ticket booking and management system with Java for the application logic and MySQL as the database.
  - Designed the user-facing pages with HTML/CSS to support booking and managing tickets.
- **SwarSanket** | Link — *Python, PyTorch, PennyLane, Faster-Whisper, React, Flutter*
  - Developed an accessible, non-invasive voice screening system for early signs of cognitive decline associated with Alzheimer's disease, available on web and mobile.
  - Users describe a standardized picture scene (Cookie Theft task) for 10–15 seconds, and the system analyzes acoustic stability, linguistic density, and structural language patterns.
  - Used a hybrid classical-quantum machine learning pipeline to compute calibrated screening indicators.
  - Built the backend in Python (PyTorch, PennyLane, Faster-Whisper), the web frontend in React, and the mobile app using Android SDK and Flutter.

Technical Skills

- **Languages**: Python, C, C++, HTML/CSS, SQL (MySQL), Java
- **AI/ML Frameworks**: PyTorch, TensorFlow
- **Tools**: Figma, NumPy, GitHub, Jupyter Notebook, VS Code
- **Research Skills**: Data Analytics, Research Documentation
- **Spoken Languages**: English, Bengali, Hindi, German

Certifications

- **Foundational Maths for Machine Learning** — 2026
  NPTEL+ Workshop, IIT Bombay — *View Certificate*
  - Covered linear algebra for machine learning. Instructor: Prof. Jayakrishnan U. Nair.
- **WADLA 4.0** — 2025
  Research Workshop on Advances in Deep Learning and Applications, IIITDM Jabalpur — *View Certificate*
  - Focused on CNNs, model compression for on-device AI, and deep learning applications.
- **Drone-based Surveillance and Image Processing** — Dec 2024 – Feb 2025
  100-Hour Training Program on aerial surveillance and image/video processing — *View Certificate*
  - Fully funded by the Ministry of Electronics and Information Technology (MeitY), Govt. of India.

Extracurricular Activities

- **Finalist at RECURSIVE Shift-8 Hackathon, SBH 2025**

Education

- **RCC Institute of Information Technology** — Kolkata, India
  Bachelor of Technology — CGPA: 9.77 — *Aug. 2024 – Present*